Climate Risk Profile

Senegal | Coastal & Agriculture Focus

40-year trends | 03 Oct 2026, 11:05 | Country Climate Profile Generator

Key Findings

+27.7%
recent-decade rainfall vs 1985-2014 baseline

62 / 190
ND-GAIN vulnerability rank, 2024

18.9 m
population (World Bank, 2025)

1. Observed trends

Mean annual temperature at the representative location (Dakar) averaged 24.3 °C over 1985-2024, with an observed warming trend of +0.13 °C per decade at this single point.[1]

Rainfall is confined to a June-September wet season peaking in August.[1] The recent decade averaged 636 mm per year against a 1985-2014 baseline of 498 mm - a +28% change - with the difference concentrated in the peak months (Figures 1-2).[1]

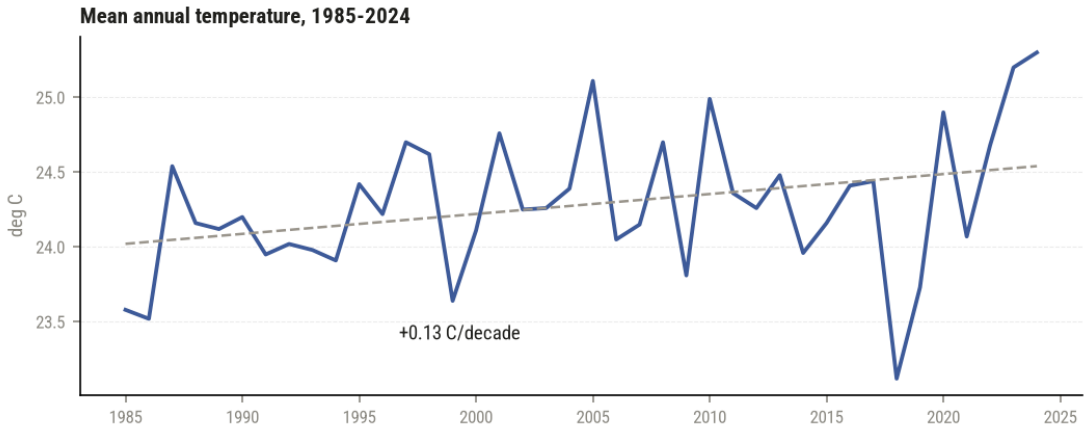


Figure 1. Mean annual temperature. Dashed line: linear trend. Single grid point (MERRA-2).

2. Seasonal rainfall pattern

Baseline vs recent decade (Figure 2).

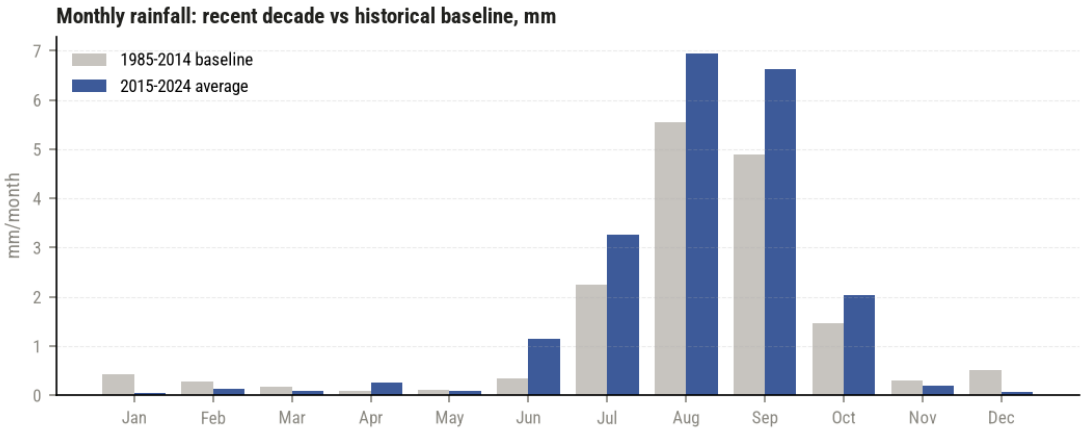


Figure 2. Monthly rainfall: baseline vs recent decade average.

3. Context and exposure

Senegal's population is about 18.9 million and growing 2.3% per year.[2] 44% of the population is rural and 20% of land is arable, keeping livelihoods tied to rain-fed agriculture and, on the coast, to fishing and urban infrastructure exposed to flooding.[2]

4. Vulnerability standing

On the ND-GAIN Country Index (2024), Senegal scores 42.3, placing it ranked 62 / 190 in vulnerability terms (a low score means high vulnerability).[3]

5. National response

Senegal's updated Nationally Determined Contribution is available through the UNFCCC registry, which lists the country's mitigation and adaptation commitments.[4]

Sources (annex)

1. NASA POWER agroclimatology API, monthly point data (-17.4, 14.7), 1985-2024, retrieved 2026-10-03 06:03 UTC. power.larc.nasa.gov
2. World Bank Open Data API, retrieved 2026-10-03 06:03 UTC. api.worldbank.org
3. ND-GAIN Country Index, 2024 scores, retrieved 2026-10-03 06:03 UTC. gain.nd.edu
4. UNFCCC NDC Registry, Senegal submission. unfccc.int/NDCREG?country=SEN